The Canonical Cut-Occurrence Shadow: A Maximal Source-Backed Partial Lift and Its Exact Boundary

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Abstract

The first unsupported object in the source-locked TPC route is a conservative, row-separated lift from every nonsoft cut path to its downstream occurrences. The existing archive does not contain those occurrences. We construct instead the strongest object that is literally determined by the archive: the canonical cut-occurrence shadow

$$S_X : \mathbb{C}^{\mathcal{C}_X^{\mathrm{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X^{\mathrm{cut}}}, \qquad S_X e_c = e_{\iota(c)}.$$

There is exactly one shadow row for every eligible-tail-open or frontier-unmapped cut path. Thus S_X is injective, row separated and exactly conservative, $\mathbf{1}^T S_X = \mathbf{1}^T$. It preserves the native tuple, prescribed shift, physical normalization, symbolic coefficient and cut multiplier, while keeping canonical-parent, stage, physical-occurrence and active-support fields absent. We prove a conditional pushforward property: every future actual occurrence lift that faithfully extends the cut data must forget to S_X. This universal property does not produce such a lift. The deterministic certificate reruns the TPC-143 source chain and contains 2988 production rows, all frontier-unmapped. A separate eligible-tail fixture is marked synthetic L0 only and never enters the production census. The result is structural L1 progress, not a fixed-shift arithmetic saving.

1 The exact input

The native, dyadic and cut archives of TPC-133–136 enumerate a literal packet through the first unsupported carrier [1–4]. Their three terminal types are

$$\mathsf{EPS}, \qquad \mathsf{ETO}, \qquad \mathsf{FUM}.$$

The already soft prefix rows EPS need no occurrence repair at this gate. The required nonsoft domain is

$$\mathcal{C}_X^{\mathrm{ns}} = \mathcal{C}_X^{\mathsf{ETO}} \sqcup \mathcal{C}_X^{\mathsf{FUM}}. \tag{1}$$

Neither summand may be omitted merely because it is empty in one finite fixture. TPC-143 attaches an explicit lift obligation to every element of $\mathcal{C}_X^{\mathrm{ns}}$, but correctly leaves every actual occurrence edge absent [6].

For $c \in \mathcal{C}_X^{\mathrm{ns}}$, let $F_{\mathrm{cut}}(c)$ denote precisely the sourced data available at the cut:

(i) the cut, upstream-path and native identities;

(ii) X, h_0, Q, U, V, the weight-source identifier and physical normalization;

(iii) the native tuple $(\ell, k, d_{\mathrm{nat}})$;

(iv) the native symbolic coefficient, native constraint witnesses and dyadic multiplier expressions; and

(v) the terminal type, block, boundary rule and formal-support status.

The subscript in d_{nat} is deliberate. No existing source identifies this divisor coordinate with a later affine intercept.

Definition 1.1 (Support namespaces)**.** The *formal support envelope* is the enumerated symbolic domain. A *canonical-parent carrier* requires a sourced inverse aggregation. The *actual active support* requires an exact nonzero coefficient certificate. We treat these as three different typed sets.

Every current nonsoft row lies in the first namespace. The other two are not inferred from a symbolic coefficient expression.

2 The cut-occurrence shadow

Definition 2.1 (Partial occurrence namespace)**.** For each $c \in \mathcal{C}_X^{\mathrm{ns}}$, define

$$\iota(c) := (c, F_{\mathrm{cut}}(c))$$

in a new typed namespace, and put

$$\mathcal{O}_X^{\mathrm{cut}} := \{\iota(c) : c \in \mathcal{C}_X^{\mathrm{ns}}\}.$$

The identifier of $\iota(c)$ is derived deterministically from the literal cut-path identifier. It is called a *partial occurrence*; it is not an actual downstream occurrence.

Definition 2.2 (Canonical shadow matrix)**.** The cut-occurrence shadow is the linear map

$$S_X : \mathbb{C}^{\mathcal{C}_X^{\mathrm{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X^{\mathrm{cut}}}, \qquad S_X e_c = e_{\iota(c)}. \tag{2}$$

Its only nonzero coefficient in column c is the exact rational number 1.

Theorem 2.3 (Exact shadow identities)**.** *The matrix S_X is total on $\mathcal{C}_X^{\mathrm{ns}}$, injective and row separated. Moreover,*

$$\mathbf{1}_{\mathcal{O}_X^{\mathrm{cut}}}^T S_X = \mathbf{1}_{\mathcal{C}_X^{\mathrm{ns}}}^T. \tag{3}$$

On every nonzero edge it intertwines the native tuple, source h_0, physical normalization, source path and formal-support status.

Proof. The rule $c \mapsto \iota(c)$ creates one distinct typed row for each literal source identifier, hence is injective and total. Each column contains one coefficient equal to 1, proving (3). The metadata statement follows because $F_{\mathrm{cut}}(c)$ is copied without reinterpretation from the corresponding source obligation. □

Remark 2.4 (Why the identity is useful but not the goal)*.* An identity matrix can be trivial algebraically and still useful as a typed boundary object. Here it gives a lossless handoff point: any later construction must explain how one partial row branches into actual rows without changing the sourced column total. It does not explain that branching.

2.1 Exact reconnection with the soft rows

Let $\mathcal{N}_X$ be the native-record set and write the conservative cut matrix as

$$M_X^{\mathrm{cut}} = \begin{pmatrix} M_X^{\mathsf{EPS}} \\ M_X^{\mathrm{ns}} \end{pmatrix}.$$

Proposition 2.5 (Shadow-level reconnection)**.** *The map $I_{\mathsf{EPS}} \oplus S_X$ satisfies*

$$\mathbf{1}_{\mathsf{EPS}}^T M_X^{\mathsf{EPS}} + \mathbf{1}_{\mathcal{O}_X^{\mathrm{cut}}}^T S_X M_X^{\mathrm{ns}} = \mathbf{1}_{\mathcal{N}_X}^T. \tag{4}$$

This is an exact coefficient identity at the cut-shadow level.

Proof. Insert (3) into the cut-column identity $\mathbf{1}_{\mathsf{EPS}}^T M_X^{\mathsf{EPS}} + \mathbf{1}_{\mathcal{C}_X^{\mathrm{ns}}}^T M_X^{\mathrm{ns}} = \mathbf{1}_{\mathcal{N}_X}^T$. □

Proposition 2.5 neither supplies physical cover nor proves that a symbolic column is nonzero. It only confirms that no cut-level coefficient was deleted by introducing the shadow.

3 Universal pushforward property

The word "maximal" below refers to sourced cut fields, not to an unknown actual carrier.

Definition 3.1 (Actual extension candidate)**.** An actual extension candidate consists of a typed row set $\mathcal{O}_X^{\mathrm{act}}$, a row-separated matrix

$$L_X : \mathbb{C}^{\mathcal{C}_X^{\mathrm{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X^{\mathrm{act}}},$$

and a forgetting map $q : \mathcal{O}_X^{\mathrm{act}} \to \mathcal{O}_X^{\mathrm{cut}}$. The map q must erase only downstream fields and must retain all fields in F_{cut}. Its linear pushforward is

$$q_* e_o = e_{q(o)}.$$

The candidate lies over the cut shadow when, for all $c, c' \in \mathcal{C}_X^{\mathrm{ns}}$,

$$\sum_{\substack{o \in \mathcal{O}_X^{\mathrm{act}} \\ q(o) = \iota(c)}} (L_X)_{o,c'} = \delta_{c,c'}. \tag{5}$$

Theorem 3.2 (Conditional universal property)**.** *Every actual extension candidate lying over the cut shadow satisfies*

$$q_* L_X = S_X. \tag{6}$$

Conversely, (6) *is equivalent to the fiber identities* (5)*.*

Proof. The coefficient of $e_{\iota(c)}$ in $q_* L_X e_{c'}$ is the left-hand side of (5). It equals $\delta_{c,c'}$, which is exactly the coefficient of $e_{\iota(c)}$ in $S_X e_{c'}$. Reading the same argument backwards proves the converse. $\square$

Corollary 3.3 (Conservation inherited from an actual lift)**.** *If $q_* L_X = S_X$, then*

$$\mathbf{1}^T_{\mathcal{O}_X^{\mathrm{cut}}} q_* L_X = \mathbf{1}^T_{\mathcal{C}_X^{\mathrm{ns}}}.$$

If, in addition, forgetting preserves the coefficient sum $\mathbf{1}^T_{\mathcal{O}_X^{\mathrm{cut}}} q_* = \mathbf{1}^T_{\mathcal{O}_X^{\mathrm{act}}}$, then $\mathbf{1}^T_{\mathcal{O}_X^{\mathrm{act}}} L_X = \mathbf{1}^T_{\mathcal{C}_X^{\mathrm{ns}}}$.

The existence of q, L_X, the number of rows in each fiber, and all actual edge coefficients remain hypotheses. Therefore theorem 3.2 is a compatibility condition for future work, not a construction of the missing lift.

4 What remains deliberately absent

The partial row reserves, but does not populate, the following field groups:

1. canonical parent, determinant fiber and Q_D edge;
2. ordered outer zero mode and Q_Z edge;
3. physical occurrence, physical group, cover and reconnection;
4. transformation-stage identifier, exact later multiplier, source and target shift tags, and downstream selector edge; and
5. the affine-corridor consumer fields and endpoint-ledger token.

The TPC-141 occurrence registry names administrative stages, but it does not give a row-level cut-to-stage-to-parent occurrence crosswalk [5]. It is therefore not used to manufacture these fields.

Proposition 4.1 (No promotion by shadow construction)**.** *Constructing S_X proves none of the following:*

(a) existence or uniqueness of an actual occurrence lift;

(b) *actual one-to-many branch multiplicities;*

(c) *determinant, zero-mode, physical-group or downstream-shift totality;*

(d) *membership in actual active support; or*

(e) *an arithmetic estimate for the nonsoft scalar.*

Proof. All five statements require fields erased by the map from a hypothetical actual extension to $\mathcal{O}_X^{\mathrm{cut}}$. The definition of S_X uses only F_{cut}, so none is a consequence of (2). □

5 Executable finite certificate

The standard-library program

`experiments/tpc153_cut_occurrence_shadow.py`

reruns the TPC-143 generator, compares its logical UTF-8/LF obligations and certificate, and emits one shadow record per source obligation. The lock mode is `CANONICAL_UTF8_LF_V2`; hashes have integrity semantics only.

For the committed production fixture,

$$\#\mathcal{C}_X^{\mathrm{ns}} = 2988, \qquad \#\mathcal{C}_X^{\mathsf{ETO}} = 0, \qquad \#\mathcal{C}_X^{\mathsf{FUM}} = 2988.$$

All 2988 source columns have one shadow row and exact column sum 1. The validator rejects deletion, duplication, terminal relabeling, nonunit weight, h_0 or normalization drift, active support promotion, an invented downstream selector, an invented actual completion, and a false L2 claim.

The production sample cannot exercise the ETO branch. A separate regression uses the TPC-135 block policy at

$$X = 2^{84}, \quad R = 2^{21}, \quad V = 2^{10}, \quad (j_L, j_K) = (38, 46), \quad D_0 = 2,$$

for which the policy classifies the block as eligible. This record is marked `SYNTHETIC_L0_ONLY`. It tests that the union (1) remains typed, but proves neither a current production ETO row nor asymptotic ETO existence.

Export	Status	Exact meaning
H1.cut_occurrence_shadow	PROVED$_{\mathrm{L1}}$	Exact total conservative partial lift.
H1.frontier_occurrence_lift	NOT-TESTABLE	Actual downstream rows and edge coefficients absent.
Current-schema-only actual-lift derivation	STOP-DECLARED-ROUTE	A shadow cannot supply erased provenance.
Selected augmented occurrence route	Open	A theorem-backed external crosswalk may extend the shadow.

6 Kill criteria and next artifact

The shadow subroute fails immediately if any nonsoft source lacks one partial row, if two source rows share a partial identifier, if a column sum differs from 1, or if native, h_0, normalization or support metadata drift. Conversely, passing these checks must not be promoted to the actual lift.

The next required artifact is a theorem-backed, row-level cut-to-stage-to-parent-to-physical-occurrence crosswalk on every ETO and FUM source. It must include exact edge multipliers, native and h_0 lineage, physical normalization, support status, cover and reconnection. The source-locked route decision remains NOT-TESTABLE until that object or the separately typed scalar-plus-ETO alternative is supplied [7].

This paper proves no frontier $o(X)$ estimate, positive fixed-h_0 L2 saving, endpoint below 1/400, prime-pair lower bound or twin-prime theorem.

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